

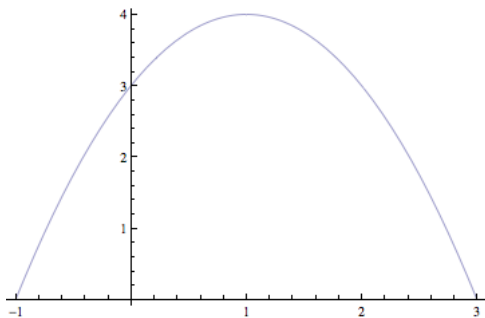
Describing a Function Using Derivatives

Thm. Let f be differentiable on (a, b) . Then the following holds:

- If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.
- If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.
- If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

Example

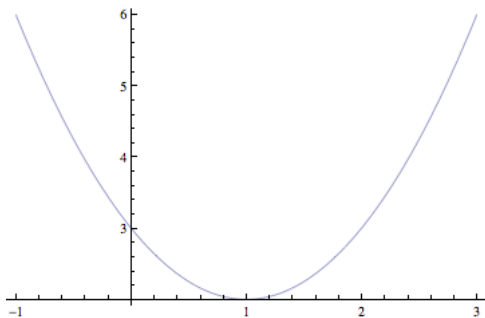
Consider the function $f(x) = -x^2 + 2x + 3$



- $f'(x) = -2x + 2$
- $f'(x) > 0$ when $x < 1$
- $f'(x) = 0$ when $x = 1$
- $f'(x) < 0$ when $x > 1$

Example 2

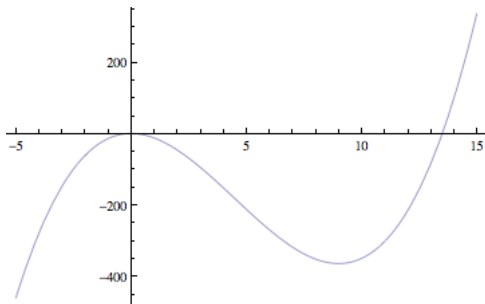
Consider the function $f(x) = x^2 - 2x + 3$



- $f'(x) = 2x - 2$
- $f'(x) < 0$ when $x < 1$
- $f'(x) = 0$ when $x = 1$
- $f'(x) > 0$ when $x > 1$

Example 3

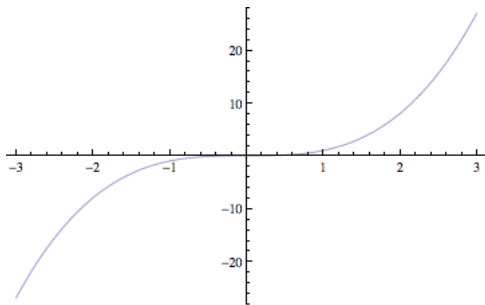
Consider the function: $f(x) = x^3 - \frac{27}{2}x^2$



- $f'(x) = 3x^2 - 27x = 3x(x - 9)$
- $f'(x) > 0$, when $x < 0$ or $x > 9$
- $f'(x) = 0$, when $x = 0$ or $x = 9$
- $f'(x) < 0$, when $x > 0$ and $x < 9$

Example 4

Consider the function $f(x) = x^3$



- $f'(x) = 3x^2$
- $f'(x) > 0$, when $x < 0$ or $x > 0$
- $f'(x) = 0$, when $x = 0$

Critical Points

- A function f has a *local maximum* at a point p if there exists $\delta > 0$ such that $f(q) \leq f(p)$ for all q such that $|p - q| < \delta$
- A function f has a *local minimum* if there exists $\delta > 0$ such that $f(q) \geq f(p)$ for all q such that $|p - q| < \delta$

Thm. Let f be a function defined on the interval $[a, b]$. If f has a local maximum at a point $x \in (a, b)$ and if $f'(x)$ exists, then $f'(x) = 0$. The same is true for a local minima.

Rolle's Theorem

Thm: Suppose that $f : [a, b] \rightarrow \mathbb{R}^1$ is continuous on $[a, b]$, and C^1 on (a, b) . If $f(a) = f(b) = 0$ then there is a point $c \in (a, b)$ such that $f'(c) = 0$. Proof is on SB pg. 824.

Mean Value Theorem

Thm. Let $f : U \rightarrow \mathbb{R}^1$ be a C^1 function on a (connected) interval in U in $\mathbb{R}^1$. For any points $a, b \in U$, there is a point c between a and b so that: $f(b) - f(a) = f'(c)(b - a)$.

Proof: Construct the function

$$g_0(x) \equiv f(b) - f(x) + \frac{f(b)-f(a)}{b-a}(x - b)$$

Note that $g_0(a) = 0$ and $g_0(b) = 0$. By Rolle's theorem, there is a point $c \in (a, b)$ so that $g'_0(c) = 0$. Differentiating (2) with respect to x yields

$$g'_0(x) = -f'(x) + \frac{(f(b)-f(a))}{b-a}$$

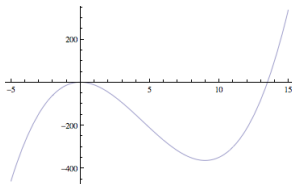
Plugging $x = c$ and $g'_0(c) = 0$ in this equation yields $f(b) - f(a) = f'(c)(b - a)$.

Mean Value Theorem

This theorem says that if you average 50 miles per hour on a trip from New York to Los Angeles, then at some point in the trip, you had to have been traveling exactly 50 miles per hour. For example if $f(x)$ is a function that assigns distance to time, x . Then the average miles per hour is given by $\frac{f(b)-f(a)}{b-a}$, which is equal to $f'(c)$, where c is some amount of time that is less than or equal to b and greater than or equal to a .

Global Maxima and Minima

Sometimes the local maxima is not equivalent to the global maxima. For example:



- The global maxima could be ∞
- If the domain for this function $f(x) = x^3 - \frac{27}{2}x^2$ has a domain of $x \in [-5, 15]$, then for what value of x does $f(x)$ obtain its global maxima?
- If the domain of the function $f(x)$ is $x \in [-5, 10]$, then for what value of x does $f(x)$ obtain its global maxima.
 - Extrema that do not occur on the boundary are called **Interior Extrema**.

Formal definitions of Boundedness

Let A be a nonempty subset of $\mathbb{R}$. The set of the upper bounds of A , denoted $U(A)$ is defined as

$$U(A) = \{u \in \mathbb{R} \mid u \geq a \forall a \in A\}$$

The set of lower bounds of A , denoted $L(A)$ is given by

$$L(A) = \{l \in \mathbb{R} \mid l \leq a \forall a \in A\}$$

If $U(A)$ is not empty, do we know how many elements are in the set $U(A)$? If $U(A)$ is empty, what can we say about the set?

What is an example?

If $U(A)$ is nonempty, then A is said to be bounded above. If $L(A)$ is non-empty A is said to be bounded below

Example of Boundedness

- If $f(x) = x$ for the domain of $x \in (0, 1)$ and we denote the range of $f(x)$ as the set A , what is the set of upper bounds $U(A)$?
- What is the set of lower bounds, $L(A)$ for the range of $f(x)$?
- If $f(x) = x$ for the domain of $x \in \mathbb{R}$, what is the set of upper bounds?
- If $f(x) = x$ for the domain of $x \in [0, 1]$, what is the set of upper bounds?

Supremum and Infimum

- If $U(A)$ is nonempty, then the supremum of A , denoted $\sup A$ is defined to be the unique point $a^* \in U(A)$ such that $a^* \leq u \forall u \in U(A)$. If $U(A)$ is empty then A has no finite upper bound and, by convention, we set $\sup A = +\infty$.
- If $L(A)$ is nonempty, then the infimum of A , denoted $\inf A$ is defined to be the unique point $\hat{a} \in L(A)$ such that $\hat{a} \geq l \forall l \in L(A)$. If $L(A)$ is empty then A has no finite lower bound and, by convention, we set $\inf A = -\infty$.

Example of Supremum, Infimum, and Extrema

- If $f(x) = x$ for the domain of $x \in (0, 1)$, what is supremum for the range of $f(x)$?
- What is the infimum for the the range of $f(x)$?
- What is the global maximum for the range of $f(x)$? What is the global minimum?
- If $f(x) = x$ for the domain of $x \in [0, 1]$ and we denote the range of $f(x)$ as the set A , what is $A \cap U(A)$?

Higher Order Derivatives

- **C^1 functions:** $\frac{d}{dx}f(x)$ is also a function of x . If $\frac{d}{dx}f(x)$ is a continuous function, we call f *continuously differentiable*. Functions that are continuously differentiable are called C^1 functions.
- **C^2 functions:** We can also differentiate $\frac{d}{dx}f(x)$, which is denoted as $\frac{d^2}{dx^2}$ or $f''(x)$ and is called the second order derivative. If the second order derivative is continuous, we say that f is twice continuously differentiable. This class of functions are called C^2 functions.
- The k th order partial derivative is denoted $\frac{d^k}{dx^k}f(x)$ or $f^{(k)}(x)$. This class of function is called C^k functions.

Notation for Higher Order Derivatives

Second order derivatives can have the following notation $f''(x)$, $\frac{d^2 f(x)}{dx^2}$, or D_x^2 . This generalize for an nth order derivative: $f^{(n)}(x)$, $\frac{d^n f(x)}{dx^n}$, or D_x^n . A higher order partial derivative is defined as ∂_x^n .

Calculating Higher Order Derivatives

To calculate a higher order derivative, just treat the lower order derivative as a function and take the derivative of this function. For example consider the function $f(x) = x^3 + 2x + 4$.

- What is the first order derivative of this function?

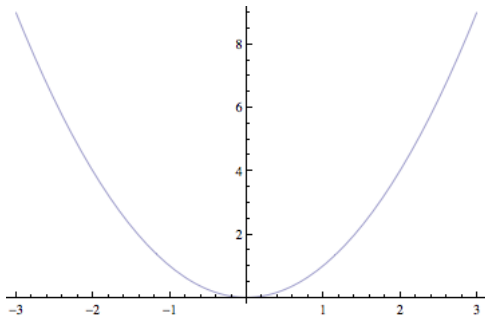
- $f'(x) = 3x^2 + 2$

- What is the second order derivative of this function?

- $f''(x) = 6x$

Concept of 2nd Order Derivative

Consider the function $f(x) = x^2$. The slope of this function at x is $f'(x) = 2x$. This slope is changing at a rate of $f''(x) = 2$. For example the slope at $f(1)$ is 2. The slope at $f(2) = 4$. This function is accelerating.



Concavity and Convexity

- **Concavity:** The function f is *concave* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- **Convexity:** The function f is *convex* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- These are different concepts than a convex set
- Concave curves have a negative second order derivative and
Convex curves have a positive second order derivative

Finding Extrema: First Derivative Test

- **Critical Points** occur when $f'(x^*) = 0$ or if $f'(x^*)$ does not exist. **Stationary points** occur when $f'(x) = 0$.
- **First Derivative Test:** To identify points that are potentially extrema, find all the points such that $f'(x^*) = 0$.
 - Consider the function $f(x) = x^2 - 8x + 3$
 - $f'(x^*) = 2x^* - 8 = 0 \implies x^* = 4$

Finding Extrema: Second Derivative Test

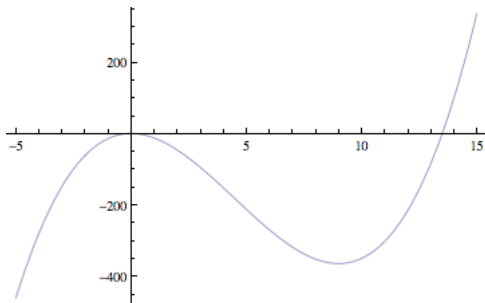
- The first derivative test identifies critical points, which are potentially extrema, but it does not tell us if the points are maxima, minima or neither.
- Second Derivative Test: If $f''(x^*) < 0$ then the point is a local maximum since the function is concave near x^* .
- If $f''(x^*) > 0$, then the point is a local minimum since the function is convex near x^* .

Inflection Points

- Not all critical points are extrema. They can also be **inflection points** or **saddle points**. For example, $f(x) = x^3$
- When $f'(x^*) = 0$ and $f''(x^*) = 0$ the second derivative test is inconclusive.
- We can verify that the point x^* is an inflection point by graphing the function
- For complicated functions we can also continue computing derivatives at the critical point x^* , and eventually we will reach a non-zero derivative.
- If the derivative is of odd order, then the point is an inflection point.
- If the derivative is of even order then point is an extrema. If the value of the non-zero derivative is negative, then the point is a maximum. If it is positive, then the point is a minimum.

Boundaries and Global Extrema

- Consider the function $f(x) = 6x + 3$ defined on the domain of $[-2, 3]$. What is $f'(x)$? What is the value of x that maximizes this function?
- Consider the function $f(x) = x^3 - \frac{27}{2}x^2$ defined on the domain of $[-5, 15]$. What are the critical points? What is the global maximum?



Extreme Value Theorem

A real-valued function that is continuous on a closed and bounded interval $[a, b]$ must attain its global maximum and minimum on that interval at least once.

- To find a global maximum and minimum
 - Check all local extrema
 - Check the boundary points
- For the function $f(x) = x^3 - \frac{27}{2}x^2$ defined on the interval of $[-5, 15]$, we know there is a local maximum at $x = 0$, which yields $f(0) = 0$. Then we only need to check the boundary points $f(-5) = -462.5$ and $f(15) = 337.5$

Taylor Polynomials

Consider a function, $f(a)$ from R^1 to R^1 . The approximation-by-differentials rule is:

$$f(a + h) \approx f(a) + f'(a)h$$

Now let $R(h; a)$ denote the difference between the two sides:

$$R(h; a) \equiv f(a + h) - f(a) - f'(a)h$$

By the definition of the derivative, $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = f'(a)$, we know $\frac{R(h;a)}{h} \rightarrow 0$ as $h \rightarrow 0$. So we know that $R(h; a)$ goes to zero and it goes to zero faster than h . *This means that the difference between the two sides of the first equation will be orders of magnitude smaller than h . We will show that with an infinitely differentiable function, Taylor polynomials can approximate the function arbitrarily well!!*

Taylor Polynomials Continued

Thm. 4 Let $f : U \rightarrow \mathbb{R}^1$ be a C^2 function defined on a (connected interval $U \in \mathbb{R}^1$). For any points a and $a + h$ in U , there exists a point c_2 between a and $a + h$ such that:

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(c_2)h^2$$

Proof see SB pg. 828

This equation is just the one-term Taylor approximation for $f(a + h) = f(a) + f'(a)h + R(h; a)$, where the remainder satisfies: $R_1(h; a) = \frac{1}{2}f''(c_2)h^2$ for some c_2 between a and $a + h$. One can easily see that $\frac{1}{2} \frac{f''(c_2)h^2}{h} \rightarrow \frac{1}{2}f''(c_2) \cdot 0 = 0$, as $h \rightarrow 0$.

Two-Term Taylor Polynomials

To get a better approximation we can use the Taylor polynomial of f of order two:

$$f(a + h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$$

Then we can define the remainder as the difference in both sides:

$$R_2(h; a) \equiv f(a + h) - f(a) - f'(a)h - \frac{1}{2}f''(a)h^2$$

Two-Term Taylor Polynomials Continued

Thm Let $f : U \rightarrow \mathbb{R}^1$ be a C^3 function defined on a (connected) interval U in $\mathbb{R}^1$. For any points a and $a + h$ in U , there exists a point c^* between a and $a + h$ such that

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2!}f''(a)h^2 + \frac{1}{3!}f'''(c^*)h^3$$

This equation is the two-term Taylor approximation where the remainder satisfies $\frac{R_2(h;a)}{h^2} = \frac{1}{3!}f'''(c^*)h^3$. Note that $\frac{R_2(h;a)}{h^2} \rightarrow 0$ as $h \rightarrow 0$ and $R_2(h;a) \rightarrow 0$ faster than $R_1(h;a) \rightarrow 0$. A two-term Taylor approximation is more precise than a one-term Taylor approximation.

General Taylor Polynomial Formula

Following the logic in the previous slide we can define a k th order Taylor polynomial as

$$\begin{aligned} f(a+h) &\approx P_k(a+h) = f(a) + f'(a)h + \frac{1}{2!}f''(a)h^2 + \dots + \frac{1}{k!}f^{[k]}(a)h^k \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} h^n \end{aligned}$$

- The Taylor Series can also be expressed as

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

- $e^x = \sum_{n=0}^{\infty} \frac{h^n}{n!} = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \dots$

- Where a , the expansion point is 0

Plotting Taylor Series

